

PRACTICAL – 12

Part-A

Code:

```
GNU nano 7.2 producer_consumer.c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <time.h>

int *buffer;
int buffer_size;
int items;

int in = 0, out = 0;
long long consumed_sum = 0;
int consumed_count = 0;

sem_t empty;
sem_t full;
pthread_mutex_t mutex;

void *producer(void *arg)
{
    for (int i = 1; i <= items; i++)
    {
        sem_wait(&empty);

        pthread_mutex_lock(&mutex);

        buffer[in] = i;
        in = (in + 1) % buffer_size;
        sem_post(&full);
    }
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ nano producer_consumer.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ nano producer_consumer.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ gcc producer_consumer.c -o producer_consumer -pthread
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ ls -l
total 28
-rw-r--r-- 1 bhardwaj bhardwaj 1 Oct 1 09:43 hello
-rwxr-xr-x 1 bhardwaj bhardwaj 17120 Oct 1 10:06 producer_consumer
-rw-r--r-- 1 bhardwaj bhardwaj 2844 Oct 1 10:05 producer_consumer.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ |
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ ./producer_consumer 2 100000
===== PRODUCER-CONSUMER RESULT =====
Buffer Size      : 2
Items Produced   : 100000
Items Consumed   : 100000
Expected Sum     : 5000050000
Consumed Sum     : 5000050000
Execution Time   : 3.551020 seconds
Throughput      : 28160.93 items/second
Synchronization : CORRECT
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ ./producer_consumer 5 100000
===== PRODUCER-CONSUMER RESULT =====
Buffer Size      : 5
Items Produced   : 100000
Items Consumed   : 100000
Expected Sum     : 5000050000
Consumed Sum     : 5000050000
Execution Time   : 1.202916 seconds
Throughput      : 83131.33 items/second
Synchronization : CORRECT
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ ./producer_consumer 10 100000
===== PRODUCER-CONSUMER RESULT =====
Buffer Size      : 10
Items Produced   : 100000
Items Consumed   : 100000
Expected Sum     : 5000050000
Consumed Sum     : 5000050000
Execution Time   : 0.527206 seconds
Throughput      : 189679.15 items/second
Synchronization : CORRECT
```

PART B — Deadlock

Code:

```
GNU nano 7.2                                deadlock_demo.c
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>

pthread_mutex_t resource1;
pthread_mutex_t resource2;

void *thread1(void *arg)
{
    printf("Thread 1: Acquiring Resource 1...\n");
    pthread_mutex_lock(&resource1);

    printf("Thread 1: Resource 1 acquired.\n");
    sleep(1);

    printf("Thread 1: Waiting for Resource 2...\n");
    pthread_mutex_lock(&resource2);

    printf("Thread 1: Resource 2 acquired.\n");

    pthread_mutex_unlock(&resource2);
    pthread_mutex_unlock(&resource1);

    return NULL;
}

[ Read 65 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  M-E Redo      M-G Copy
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ gcc deadlock_demo.c -o deadlock_demo -pthread
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ timeout 5s ./deadlock_demo
===== DEADLOCK DEMONSTRATION =====
```

```
Thread 2: Acquiring Resource 2...
Thread 2: Resource 2 acquired.
Thread 1: Acquiring Resource 1...
Thread 1: Resource 1 acquired.
Thread 1: Waiting for Resource 2...
Thread 2: Waiting for Resource 1...
```

```
Deadlock prevented using resource ordering.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ ls -lh
total 76K
-rwxr-xr-x 1 bhardwaj bhardwaj 17K Oct  1 10:09 deadlock_demo
-rw-r--r-- 1 bhardwaj bhardwaj 1.4K Oct  1 10:08 deadlock_demo.c
-rwxr-xr-x 1 bhardwaj bhardwaj 17K Oct  1 12:31 deadlock_prevention
-rw-r--r-- 1 bhardwaj bhardwaj 1.7K Oct  1 12:30 deadlock_prevention.c
-rw-r--r-- 1 bhardwaj bhardwaj  1 Oct  1 09:43 hello
-rwxr-xr-x 1 bhardwaj bhardwaj 17K Oct  1 10:06 producer_consumer
-rw-r--r-- 1 bhardwaj bhardwaj 2.8K Oct  1 10:05 producer_consumer.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-12$ |
```